/***

PROGRAM NAME: ADEG.SAS
PROJECT : ADAM ECG ANALYSIS DATASET (ADEG)
PROGRAMMER: NITHISH M S
DATE : 09JUN2026

PURPOSE:
CREATE ADEG DATASET FROM EG AND SUPPEG DATASETS AND GENERATE XPT FILE FOR SUBMISSION.
***/

/* PROJECT ON ECG-EG */

PROC IMPORT DATAFILE="/HOME/U64168920/ADAM_POJ/PROJECT ADAM-BDS EG RAW DATA.XLSX"
OUT=EG
DBMS=XLSX REPLACE;
GETNAMES=YES;
RUN;

Table: WORK.EG View: Column names Filter: (none)

Total rows: 45 Total columns: 15 Rows 1-45

STUDYID	DOMAIN	USUBJID	VISIT	VISITNUM	EGDTC	EG
SCL002	EG	SCL002-002-03	VISIT 1	10	2020-09-30T11:40:00	EC
SCL002	EG	SCL002-002-03	VISIT 1	10	2020-09-30T11:40:00	Re
SCL002	EG	SCL002-002-03	VISIT 1	10	2020-09-30T11:40:00	Su
SCL002	EG	SCL002-002-03	VISIT 1	10	2020-09-30T11:40:00	Su
SCL002	EG	SCL002-002-03	VISIT 1	10	2020-09-30T11:40:00	QR
SCL002	EG	SCL002-002-03	VISIT 1	10	2020-09-30T11:40:00	QT
SCL002	EG	SCL002-002-03	VISIT 1	10	2020-09-30T11:40:00	Su
SCL002	EG	SCL002-002-03	VISIT 1	10	2020-09-30T11:40:00	Su
SCL002	EG	SCL002-005-02	VISIT 3	30	2020-10-14T7:45:00	EC
SCL002	EG	SCL002-005-02	VISIT 3	30	2020-10-14T7:45:00	Re
SCL002	EG	SCL002-005-02	VISIT 3	30	2020-10-14T7:45:00	Su
SCL002	EG	SCL002-005-02	VISIT 3	30	2020-10-14T7:45:00	Su
SCL002	EG	SCL002-005-02	VISIT 3	30	2020-10-14T7:45:00	QR
SCL002	EG	SCL002-005-02	VISIT 3	30	2020-10-14T7:45:00	QT
SCL002	EG	SCL002-005-02	VISIT 3	30	2020-10-14T7:45:00	Su

DATA EG;
SET WORK.EG;
RUN;

```
PROC IMPORT DATAFILE="/HOME/U64168920/ADAM_POJ/PROJECT ADAM-BDS
SUPPADAE RAW DATA.XLSX"
OUT=SUPPEG
DBMS=XLSX REPLACE;
GETNAMES=YES;
RUN;
```

Table: WORK.SUPPEG View: Column names Filter: (none)

Total rows: 44 Total columns: 10 Rows 1-44

	studyid	usubjid	rdomain	idvar	idvarval	qnam	qlabel	qval
	SCL002	SCL002-002-03	EG	EGSEQ	5	SPEC	Specify	T ABNORM
	SCL002	SCL002-002-03	EG	EGSEQ	7	SPEC	Specify	T ABNORM
	SCL002	SCL002-002-03	EG	EGSEQ	8	SPEC	Specify	T ABNORM
	SCL002	SCL002-002-03	EG	EGSEQ	3	SPEC	Specify	T ABNORM
	SCL002	SCL002-002-03	EG	EGSEQ	1	SPEC	Specify	T ABNORM
	SCL002	SCL002-002-03	EG	EGSEQ	2	SPEC	Specify	T ABNORM
	SCL002	SCL002-002-03	EG	EGSEQ	6	SPEC	Specify	T ABNORM
	SCL002	SCL002-002-03	EG	EGSEQ	4	SPEC	Specify	T ABNORM
	SCL002	SCL002-005-02	EG	EGSEQ	14	SPEC	Specify	LEFT ATRIA
0	SCL002	SCL002-005-02	EG	EGSEQ	11	SPEC	Specify	LEFT ATRIA
1	SCL002	SCL002-005-02	EG	EGSEQ	15	SPEC	Specify	LEFT ATRIA
2	SCL002	SCL002-005-02	EG	EGSEQ	10	SPEC	Specify	LEFT ATRIA
3	SCL002	SCL002-005-02	EG	EGSEQ	12	SPEC	Specify	LEFT ATRIA
4	SCL002	SCL002-005-02	EG	EGSEQ	9	SPEC	Specify	LEFT ATRIA
5	SCL002	SCL002-005-02	EG	EGSEQ	22	SPEC	Specify	LEFT ATRIA
6	SCL002	SCL002-005-02	EG	EGSEQ	21	SPEC	Specify	LEFT ATRIA
7	SCL002	SCL002-005-02	EG	EGSEQ	11	SPEC	Specify	LEFT ATRIA
8	SCL002	SCL002-005-02	EG	EGSEQ	10	SPEC	Specify	LEFT ATRIA

```
DATA SUPPEG;
SET WORK.SUPPEG;
RUN;

/* MERGE BOTH THE DATASET */
PROC SORT DATA=EG OUT=EG1;BY USUBJID EGSEQ;RUN;

PROC SORT DATA=SUPPEG OUT=SUPPEG1;BY USUBJID EGSEQ;RUN;

DATA ADEG;
MERGE EG1(IN=A) SUPPEG1(IN=B);
BY USUBJID EGSEQ;
```

```
IF A AND B;
RUN;
```

Table of Contents

Obs	STUDYID	DOMAIN	USUBJID	VISIT	VISITNUM	EGDTC	EGTEST	EGTESTCD	EGCAT	EGORRES	EGSTRESC	EGSTRESN	EGBLFL	EGSEQ	O	rdomain	idvar	idvarval	qnam	qlabel	qval	qorig
1	SCL002	EG	SCL002-002-03	VISIT 1	10	2020-09-30T11:40:00	ECG abnormalty 1	ABN1_S	FINDING	INCMPLT RT B	INCMPLT RT B	.	Y	1		EG	EGSEQ	1	SPEC	Specify	T ABNORMALITY IN ANTEROSEPTAL LEADS	CRF
2	SCL002	EG	SCL002-002-03	VISIT 1	10	2020-09-30T11:40:00	Result of ECG	ECGR_S	FINDING	ABNORMAL, NO	ABNORMAL, NO	.	Y	2		EG	EGSEQ	2	SPEC	Specify	T ABNORMALITY IN ANTEROSEPTAL LEADS	CRF
3	SCL002	EG	SCL002-002-03	VISIT 1	10	2020-09-30T11:40:00	Summary (Mean) Heart Rate	HRMEAN	MEASUREMENT	62	62	62	Y	3		EG	EGSEQ	3	SPEC	Specify	T ABNORMALITY IN ANTEROSEPTAL LEADS	CRF
4	SCL002	EG	SCL002-002-03	VISIT 1	10	2020-09-30T11:40:00	Summary (Mean) PR Duration	PRMEAN	MEASUREMENT	170	170	170	Y	4		EG	EGSEQ	4	SPEC	Specify	T ABNORMALITY IN ANTEROSEPTAL LEADS	CRF
5	SCL002	EG	SCL002-002-03	VISIT 1	10	2020-09-30T11:40:00	QRS-Duration	QRSDUR	MEASUREMENT	86	86	86	Y	5		EG	EGSEQ	5	SPEC	Specify	T ABNORMALITY IN ANTEROSEPTAL LEADS	CRF
6	SCL002	EG	SCL002-002-03	VISIT 1	10	2020-09-30T11:40:00	QTcB-Bazett's Correction Formula	QTCB	MEASUREMENT	402	402	402	Y	6		EG	EGSEQ	6	SPEC	Specify	T ABNORMALITY IN ANTEROSEPTAL LEADS	CRF
7	SCL002	EG	SCL002-002-03	VISIT 1	10	2020-09-30T11:40:00	Summary (Mean) QT Duration	QTMEAN	MEASUREMENT	396	396	396	Y	7		EG	EGSEQ	7	SPEC	Specify	T ABNORMALITY IN ANTEROSEPTAL LEADS	CRF
8	SCL002	EG	SCL002-002-03	VISIT 1	10	2020-09-30T11:40:00	Summary (Mean) RR Duration	RRMEAN	MEASUREMENT	967	967	967	Y	8		EG	EGSEQ	8	SPEC	Specify	T ABNORMALITY IN ANTEROSEPTAL LEADS	CRF
9	SCL002	EG	SCL002-005-02	VISIT 3	30	2020-10-14T7:45:00	ECG abnormalty 1	ABN1_S	FINDING	INCMPLT RT B	INCMPLT RT B	.		9		EG	EGSEQ	9	SPEC	Specify	T ABNORMALITY IN ANTEROSEPTAL LEADS	CRF
10	SCL002	EG	SCL002-005-02	FINAL VISIT	40	2020-10-21T10:26:00	Summary (Mean) RR Duration	RRMEAN	MEASUREMENT	882	882	882		9		EG	EGSEQ	9	SPEC	Specify	LEFT ATRIAL ENLARGEMENT RIGHT VENTRICULAR CONDUCTION DELAY	CRF

```
DATA ADEG1;
SET ADEG;
AVAL=INPUT(EGSTRESN,BEST.);
ABLFL=EGBLFL;
PARAM=EGTEST;

IF LENGTH(EGDTC) > 10 THEN DO;
ADT=INPUT(SUBSTR(EGDTC,1,10),YYMMDD10.);
ATM=INPUT(SUBSTR(EGDTC,12),TIME8.);
END;
ELSE DO;
ADT=INPUT(SUBSTR(EGDTC,1,10),YYMMDD10.);
END;
FORMAT ADT DATE9. ATM TIME8.;
AVISITN=VISITNUM/10;
AVISIT=VISIT;
RUN;
```

Table of Contents

Obs	STUDYID	DOMAIN	USUBJID	VISIT	VISITNUM	EGDTC	EGTEST	EGTESTCD	EGCAT	EGORRES	EGSTRESC	EGSTRESN	EGBLFL	EGSEQ	O	rdom
1	SCL002	EG	SCL002-002-03	VISIT 1	10	2020-09-30T11:40:00	ECG abnormalty 1	ABN1_S	FINDING	INCMPLT RT B	INCMPLT RT B	.	Y	1		EG
2	SCL002	EG	SCL002-002-03	VISIT 1	10	2020-09-30T11:40:00	Result of ECG	ECGR_S	FINDING	ABNORMAL, NO	ABNORMAL, NO	.	Y	2		EG
3	SCL002	EG	SCL002-002-03	VISIT 1	10	2020-09-30T11:40:00	Summary (Mean) Heart Rate	HRMEAN	MEASUREMENT	62	62	62	Y	3		EG
4	SCL002	EG	SCL002-002-03	VISIT 1	10	2020-09-30T11:40:00	Summary (Mean) PR Duration	PRMEAN	MEASUREMENT	170	170	170	Y	4		EG
5	SCL002	EG	SCL002-002-03	VISIT 1	10	2020-09-30T11:40:00	QRS-Duration	QRSDUR	MEASUREMENT	86	86	86	Y	5		EG
6	SCL002	EG	SCL002-002-03	VISIT 1	10	2020-09-30T11:40:00	QTcB-Bazett's Correction Formula	QTCB	MEASUREMENT	402	402	402	Y	6		EG
7	SCL002	EG	SCL002-002-03	VISIT 1	10	2020-09-30T11:40:00	Summary (Mean) QT Duration	QTMEAN	MEASUREMENT	396	396	396	Y	7		EG
8	SCL002	EG	SCL002-	VISIT	10	2020-09-	Summary	RRMEAN	MEASUREMENT	967	967	967	Y	8		EG

```
DATA ADEG1;
SET ADEG1;
WHERE ABLFL="Y";
KEEP EGBLFL EGSTRESN VISIT;
RUN;

DATA ADEG2;
SET ADEG1;
RETAIN BASE;
IF ABLFL="Y" THEN DO;
BASE=AVAL;
END;
RUN;
```

Table: WORK.ADEG1 View: Column names

Total rows: 16 Total columns: 3

	VISIT	EGSTRESN	EGBLFL
1	VISIT 1	.	Y
2	VISIT 1	.	Y
3	VISIT 1	62	Y
4	VISIT 1	170	Y
5	VISIT 1	86	Y
6	VISIT 1	402	Y
7	VISIT 1	396	Y
8	VISIT 1	967	Y
9	FINAL VISIT	.	Y
10	VISIT 1	.	Y
11	VISIT 1	60	Y
12	VISIT 1	168	Y
13	VISIT 1	78	Y
14	VISIT 1	444	Y
15	VISIT 1	426	Y
16	VISIT 1	1000	Y

```
DATA ADEG3;
SET ADEG2;
IF AVAL NE . AND BASE NE . THEN CHG=AVAL-BASE;
IF CHG NE . AND NOT IN(.,0) THEN PCHG=(CHG*100)/BASE;
KEEP
STUDYID
USUBJID
EGEVAL
EGCAT
AVAL
AVALC
ABLFL
PARAM
PARAMCD
ADT
ATM
AVISITN
AVISIT
```

```
BASE
CHG
PCHG;
RUN;
```

```
PROC PRINT DATA=ADEG3 (FIRSTOBS=1 OBS=10);
RUN;
```

Obs	STUDYID	USUBJID	EGCAT	AVAL	ABLFL	PARAM	ADT	ATM	AVISITN	AVISIT	BASE	CHG	PCHG
1	SCL002	SCL002-002-03	FINDING	.	Y	ECG abnormalty 1	30SEP2020	11:40:00	1	VISIT 1	.	.	.
2	SCL002	SCL002-002-03	FINDING	.	Y	Result of ECG	30SEP2020	11:40:00	1	VISIT 1	.	.	.
3	SCL002	SCL002-002-03	MEASUREMENT	62	Y	Summary (Mean) Heart Rate	30SEP2020	11:40:00	1	VISIT 1	62	0	0.00000
4	SCL002	SCL002-002-03	MEASUREMENT	170	Y	Summary (Mean) PR Duration	30SEP2020	11:40:00	1	VISIT 1	170	0	0.00000
5	SCL002	SCL002-002-03	MEASUREMENT	86	Y	QRS-Duration	30SEP2020	11:40:00	1	VISIT 1	86	0	0.00000
6	SCL002	SCL002-002-03	MEASUREMENT	402	Y	QTcB-Bazett's Correction Formula	30SEP2020	11:40:00	1	VISIT 1	402	0	0.00000
7	SCL002	SCL002-002-03	MEASUREMENT	396	Y	Summary (Mean) QT Duration	30SEP2020	11:40:00	1	VISIT 1	396	0	0.00000
8	SCL002	SCL002-002-03	MEASUREMENT	967	Y	Summary (Mean) RR Duration	30SEP2020	11:40:00	1	VISIT 1	967	0	0.00000
9	SCL002	SCL002-005-02	FINDING	.		ECG abnormalty 1	14OCT2020	7:45:00	3	VISIT 3	967	.	.
10	SCL002	SCL002-005-02	MEASUREMENT	882		Summary (Mean) RR Duration	21OCT2020	10:26:00	4	FINAL VISIT	967	-85	-8.79007

```
/* LABEL */
DATA FINAL;
LENGTH
STUDYID $25
USUBJID $25
EGEVAL  $200
EGCAT   $50
AVALC   $200
ABLFL   $1
PARAM   $200
PARAMCD $20
AVISIT  $100;
SET ADEG3;
LABEL
STUDYID = "STUDY IDENTIFIER"
USUBJID = "UNIQUE SUBJECT IDENTIFIER"
EGEVAL  = "ECG EVALUATION RESULT"
EGCAT   = "ECG CATEGORY"
AVAL    = "ANALYSIS VALUE"
AVALC   = "ANALYSIS VALUE (CHARACTER)"
ABLFL   = "BASELINE RECORD FLAG"
PARAM   = "PARAMETER"
PARAMCD = "PARAMETER CODE"
ADT     = "ANALYSIS DATE"
ATM     = "ANALYSIS TIME"
AVISITN = "ANALYSIS VISIT (N)"
AVISIT  = "ANALYSIS VISIT"
```

```
BASE   = "BASELINE VALUE"
CHG    = "CHANGE FROM BASELINE"
PCHG   = "PERCENT CHANGE FROM BASELINE";
RUN;
```

SAS Table Properties

General | Columns | Extended Attributes | Column Extended Attributes

Column Name	Type	Length	Format	Informat	Label
STUDYID	Char	25	$6.	$6.	Study Identifier
USUBJID	Char	25	$13.	$13.	Unique Subject Identifier
EGEVAL	Char	200			ECG Evaluation Result
EGCAT	Char	50	$11.	$11.	ECG Category
AVALC	Char	200			Analysis Value (Character)
ABLFL	Char	1			Baseline Record Flag
PARAM	Char	200			Parameter

```
/* XPT OUT */
LIBNAME XPTOUT XPORT "/HOME/U64168920/ADAM_POJ/EG.XPT";

DATA XPTOUT.FINAL;
SET FINAL;
RUN;

LIBNAME XPTOUT CLEAR;
```

```
1          OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
68
69         LIBNAME XPTOUT XPORT "/home/u64168920/ADaM_poj/EG.xpt";
NOTE: Libref XPTOUT was successfully assigned as follows:
      Engine:        XPORT
      Physical Name: /home/u64168920/ADaM_poj/EG.xpt
70
71         DATA XPTOUT.FINAL;
72         SET FINAL;
73         RUN;

NOTE: There were 44 observations read from the data set WORK.FINAL.
NOTE: The data set XPTOUT.FINAL has 44 observations and 16 variables.
NOTE: DATA statement used (Total process time):
```